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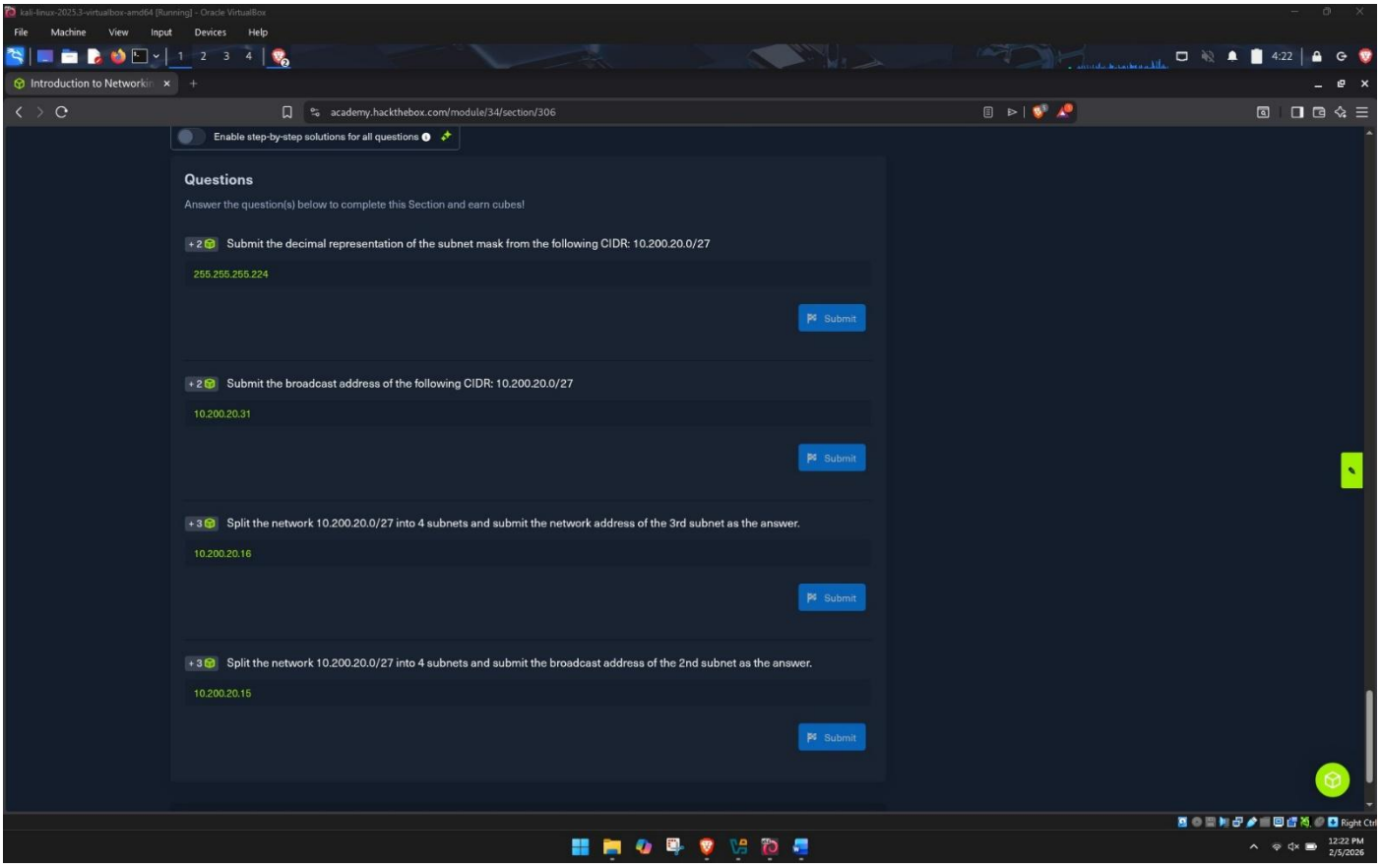
# Introduction

This module introduced me to the concept of networks. I discovered that networks rely on various topologies and protocols to enable computer communication. I found out that flat networks are operationally reliable but offer very little security. This module revealed that segmenting large networks into smaller ones creates vital layers of defence. I also learned that these structural layers effectively slow down attackers by making silent pivoting much more difficult.

# Subnetting

This section focuses on the splitting of an ipv4 address into multiple smaller segments, which is subnetting.



1. The first question prompted for the decimal representation of the subnet mask from the CIDR 10.200.20.0 /27

In CIDR notation, /27 means that the 27 bits are ‘1’s in the subnet mask.

An IPv4 address usually has 32 bits in total. Therefore:

32-27 = 5 bits and therefore:

27 bits for network and 5 bits for hosts.

Then convert the subnet mask in binary.

An IPv4 address has 4 octets, each with 8 bits.

11111111.11111111.11111111.00000000

Now, we need 27 ones.

For the first 3 octets (24 bits), each full octet is 8 bits and all are 1s.

We have 27 bits.

Therefore, 27 – 24 = 3

Therefore, the first 3 octets will be ones and the fourth will have three ones in it.

11111111.11111111.11111111.11100000

This is the subnet mask in binary. In decimal,

Bit value increases from one in multiples of two from the right going to the left.

Therefore, the first 3 octets are as such:

128+64+32+16+8+4+2+1 = 255

And the last octet will be as such:

128+64+32 = 224

Therefore, the subnet mask will be:

255.255.255.224

2. The second question prompted for the broadcast address of the same address.

   Now we know for the first question, the host bits were: 32-27 = 5 host bits.

   The number of addresses are in the subnet are 2^5 = 32; and therefore, each subnet has 32 addresses.

   The last octet in decimal is 224. Therefore:

   256 – 224 = 32, and therefore, subnets increase by 32 in the last octet. This is the block size.

   When we list the subnet ranges, it will look like this:

| Subnet | Network Address | Broadcast Address |
|---|---|---|
| 1 | .0 | .31 |
| 2 | .32 | .63 |
| 3 | .64 | .95 |
| 4 | .96 | .127 |
| 5 | .128 | .159 |
| 6 | .160 | .191 |
| 7 | .192 | .223 |
| 8 | .224 | .255 |

   Then, we identify the subnet we're in. 10.200.20.0. This starts at 0.

   Therefore, it is the first subnet = 10.200.20.31

3. The third question had to do with splitting the network 10.200.20.0/27 into 4 subnets and submit the network address of the 3rd subnet as the answer.

   The goal is to split 10.200.20.0/27 into 4 subnets and find the network address of the 3rd subnet.

   First, we have to borrow bits. To get 4 subnets, we solve 2^n = 4, which means we borrow 2 bits.

   The original prefix is /27. Adding the 2 borrowed bits gives a new prefix of /29 (Subnet Mask: 255.255.255.248).

We then have to find the Increment: The increment is determined by the last bit of the mask or 2^ {32-29} = 2^3 = {8}.

We then have to step through the subnets.

Subnet 1 = 10.200.20.0

Subnet 2 = 10.200.20.0 + 8 = 10.200.20.8

Subnet 3 = 10.200.20.8 + 8 = 10.200.20.16

Therefore,

**Answer = 10.200.20.16**

4. The last question had to do with splitting the same network but this time submitting the broadcast address of the second subnet as the answer.

   The goal is to split 10.200.20.0/27 into 4 subnets and find the broadcast address of the 2nd subnet.

   First, we have to borrow bits and increment. Using the same logic as above, we use a /29 prefix with an increment of 8.

   We have to find the Network Addresses

   Subnet 1 = 10.200.20.0

   Subnet 2 = 10.200.20.8

   Subnet 3 = 10.200.20.16

   We then determine the Broadcast. The broadcast address of any subnet is always one address before the next subnet's network address.

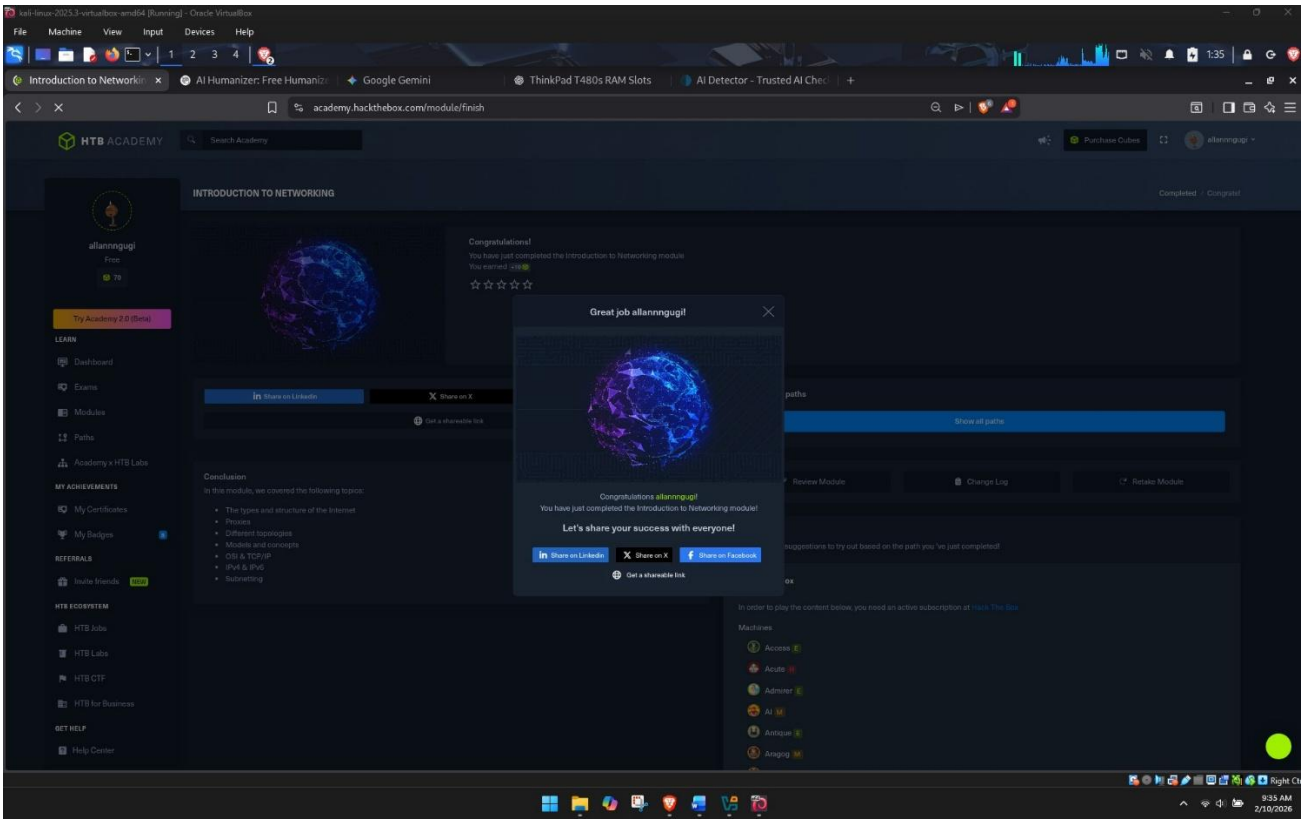
   Broadcast of Subnet 2 = (Network Address of Subnet 3) – 1

   Broadcast of Subnet 2 = 10.200.20.16 - 1 = 10.200.20.15

   Therefore,

   **Answer = 10.200.20.15**

# Screenshot Showing Module Completion




# Conclusion

The module taught all fundamental networking concepts, which included basic network structures and topologies and advanced IP address systems, which included IPv4 and IPv6. The OSI and TCP/IP models showed me how security protocols and key exchange methods create secure connections. I have learned the necessary network defence terms and technical skills through my study of the module.

# Link Showing Module Completion

https://academy.hackthebox.com/achievement/2384006/34